

Ashutosh Kumar

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Professional Summary

Senior AI/ML Engineer with 4+ years of end-to-end ownership building and deploying **Generative AI**, **Large Language Model (LLM)**, and **Agentic AI** systems at enterprise scale. Deep hands-on expertise spanning **Azure OpenAI**, **LangChain**, **LangGraph**, **Retrieval-Augmented Generation (RAG)**, **Model Context Protocol (MCP)**, **LoRA/PEFT fine-tuning**, **Responsible AI**, and **MLOps** – with production impact across regulated **banking** and safety-critical **autonomous driving** domains. Consistent record of shipping reliable, compliant, and high-impact AI systems.

Experience

Senior Software Engineer – Generative AI & Applied ML

Sep 2025 – Present

Photon

Bengaluru, India

- Architected a **multi-tenant enterprise GenAI platform** on **Azure OpenAI (GPT-4, LLaMA)**, within **Photon AI**, enabling scalable knowledge automation workflows across global client organisations.
- Designed production **Agentic AI** systems with **LangChain + LangGraph**: multi-step reasoning, dynamic tool-use, memory management, and prompt routing across multi-agent pipelines.
- Built **RAG** pipelines integrating **Azure Cosmos DB** with vector search (**FAISS, Pinecone**); reduced hallucination and bias by **20%** via **Responsible AI** guardrails and continuous evaluation.
- Fine-tuned domain LLMs using **LoRA / QLoRA (PEFT)**; boosted task success rate and response relevance by **35%**.
- Owned **MLOps**: containerised deployments (**Docker**), **CI/CD**, and **MLflow** tracking – reducing time-to-production and improving release reliability.

Senior Engineer – AI/ML & Generative AI R&D

Nov 2021 – July 2025

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Bengaluru, India

- Architected scalable **ML validation frameworks** for perception and prediction models in autonomous driving.
- Built **synthetic data generation** pipelines via GenAI, expanding rare edge-case coverage and boosting **F1-score by 15%** on production ADAS datasets.
- Trained **CNN** and **LSTM** models on fused **LiDAR, radar, and camera** data; reduced false positives by **25%** through systematic A/B evaluation.
- Productionised pipelines with **Docker, MLflow, CI/CD**; improved predictive accuracy and cut validation cycle time by **20%**.

Key Projects

Conversational AI Studio for Banking – Client: Temenos

Azure OpenAI · LangChain · LangGraph · MCP · RAG · FastAPI · Cosmos DB · Harness

- **Challenge:** Enterprise banking teams lacked a rapid prototyping platform to convert business requirements into production-ready mobile and web applications, resulting in months-long development cycles and high delivery costs.
- **Solution:** Built a **banking-grade Conversational AI Studio** for regulated financial apps, using **Azure OpenAI + LangGraph + MCP Servers + Temenos Fabric API** that converts natural language queries into complete banking applications, auto-generating **React and Kotlin** code for web and mobile, with **AML/KYC-compliant** data flows, audit-ready API contracts, and a **visual editor** with **Responsible AI** guardrails for bias checks and PII redaction.
- **Impact:** Cut design-to-production time by **60%**, requirement accuracy by **25%**; live across multiple Temenos banking clients.

Autonomous RoboTaxi – Predictive Modeling & GenAI – Client: ZOOX (Amazon)

PyTorch · AWS SageMaker · Amazon Bedrock · Lambda · LLaMA · NVIDIA Omniverse

- **Challenge:** AV prediction models failed in rare safety-critical edge cases due to limited real-world data, causing high false-positive rates that blocked production deployment.
- **Solution & Impact:** Developed **multi-sensor fusion** (camera, LiDAR, radar) trajectory prediction in **PyTorch**; built **GenAI-driven synthetic data pipelines** in NVIDIA Omniverse expanding edge-case coverage by **35%**; integrated **RAG analytics on AWS Bedrock (LLaMA)** improving simulation efficiency and model robustness by **20%**.

Technical Skills

Languages: Python, SQL, C/C++, Data Structures & Algorithms

Generative AI & LLMs: Azure OpenAI, GPT-4, GPT-5, Claude Sonnet, Gemini API, LLaMA, Hugging Face, Agentic AI, LangChain, LangGraph, RAG, MCP, Responsible AI, Prompt Engineering, LoRA, QLoRA, PEFT, LLMops, AIops

ML / Deep Learning: PyTorch, TensorFlow, Scikit-learn, OpenCV, FAISS, Pinecone, MLflow

Cloud & MLOps: Azure, AWS (SageMaker, Bedrock, Lambda), GCP, Vertex AI, Docker, Kubernetes, Terraform, CI/CD, FastAPI

Data & Analytics: Cosmos DB, NumPy, Pandas, Power BI, Looker, Git, GitHub, Linux, JIRA, Confluence

Education

University of Engineering and Management (UEM)

Kolkata, India

B.Tech, Computer Science & Engineering – CGPA: 8.6 / 10

2017 – 2021

Certifications & Awards

Model Context Protocol: Advanced Topics · Claude Code in Action – Anthropic

Mar 2026

Transformer-Based NLP · Generative AI with Diffusion Models – NVIDIA

Aug 2025

Awards: *Extra Mile Award* – ADAS predictive accuracy +40% · *Project Excellence Award* – ML model stability & on-time delivery